Climate change: impact on the agriculture in Ukraine

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Ukraine, the second largest country in Europe, is known as the region's breadbasket thanks to its black "chernozem" soil, which is highly fertile and rich in organic matter called humus. Agriculture is a crucial sector for the Ukrainian economy and contributed on average 10 % to gross domestic product between 2005 and 2012. Within agriculture, cereal production contributes about one third to total production value and is the most important land use in terms of area used. Overall, cereal production amounted to 42 million tons on average between 2005 and 2012.

Environmental zones of Ukraine

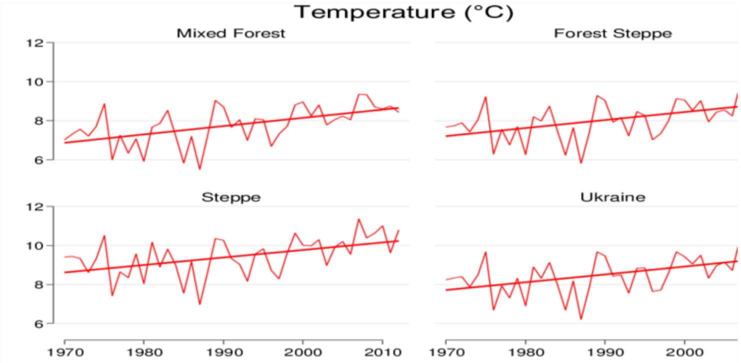
Ukrainian agriculture might experience positive outcomes from climate change in some regions because of increasing winter temperatures and winter precipitation, a longer frost - free season, and higher CO^2 concentration. Suitable croplands may expand in response to projected climate change, particularly in northern Ukraine, but the projections are subject to high uncertainty owing to unknown responses of agricultural land use with respect to the CO^2- fertilization effects on yields and unforeseeable adaptation developments.

However, prolonged periods with summer temperatures far above the long - term mean may become the norm, particularly in the southern Steppe zone of Ukraine, threatening crop production during a key period for crop development in this agriculturally most productive region with its widespread, fertile Chernozem soils. Unfortunately, detailed assessments of how different scenarios of future climate change may affect of agricultural products in Ukraine are still missing.

Past 15 years, drought events in Ukraine have been increasing – both in intensity and in frequency – due largely to a changing climate. Droughts now occur on average once every three years, causing a significant decline in crop productivity. The impacts are particularly felt by highly productive areas of the country, such as the Steppe area in the south which currently produces 50% of the grain for Ukraine.

While climatic conditions are generally favourable in Ukraine, climatic variability – which is expected to increase with climate change – is a considerable risk for agriculture.

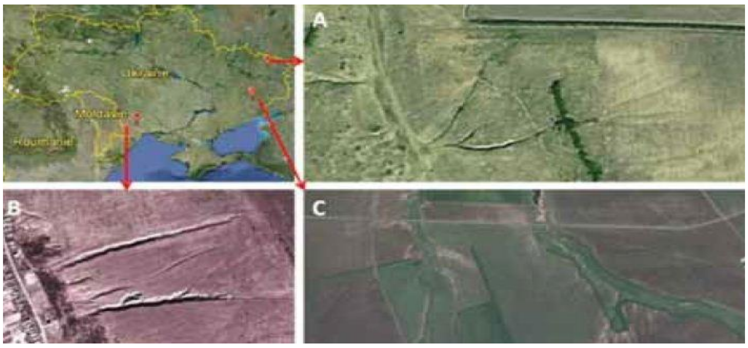
It is expected that climate change will further exacerbate the already high volatility of agricultural production and negatively affect food security.



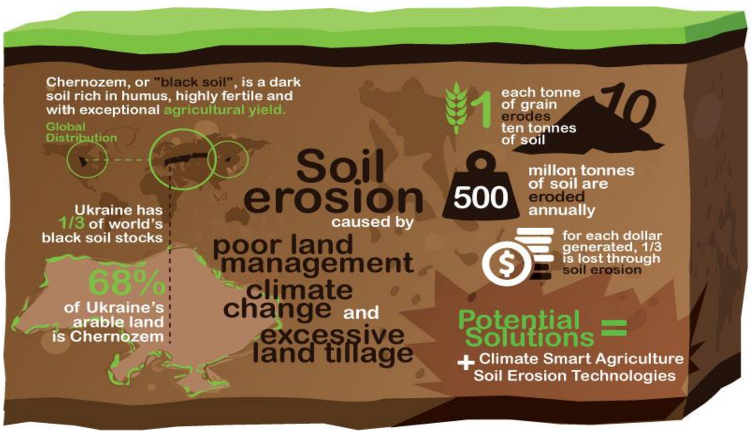
Climate variations in Ukraine, 1970 to 2012

Despite the favorable conditions, however, a major challenge for Ukraine is soil erosion. Over the decades, chernozem soil across the country has been increasingly degraded by poor land management and subsequent soil erosion. It is estimated that over 500 million tonnes of soil are eroded annually from arable land in Ukraine, resulting in a loss of fertility in over 32 thousand hectares of soil.

The detrimental impacts on crop production and the economy in Ukraine cannot be overstated: for each dollar of added agricultural value generated, one-third is lost through erosion – with ten tonnes of soil eroded for each tonne of grain produced.



Soil erosion in Ukraine is visible from satellites. (Source: Google Earth)



There is strong evidence that soil erosion in Ukraine is accelerating, leading to major natural damage in other ways too, such as siltation of rivers, harbors, and dam reservoirs (which feed hydroelectric power stations). But, the problem is more acute in some areas than others: in the south-east of the country, for example, soil has been eroded to the extent of desertification.

At the national level, for example, conservation agriculture on 17 million hectares of land could help generate income of US$ 4.4 billion – equivalent to 34 percent of agricultural GDP. Without counting global environmental and food security benefits, this outcome would almost counterbalance the depletion in natural capital of US$ 5 billion caused by soil erosion in Ukraine.

Climate Smart Agriculture

Excessive land tillage is known to be a major driver of soil erosion. The practice of conservation agriculture, on the other hand, is seen as a sustainable and effective alternative. No-till conservation agriculture reduces soil erosion, maintains soil fertility, enhances drought resilience and significantly reduces production costs by minimizing fuel consumption.

Large scale adoption of climate smart agriculture such as conservation agriculture, combined with effective soil erosion technologies, can potentially have significant benefits for Ukraine, and indeed for the world.